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Fostering farmers' innovativeness through training and extension approaches: A systematic literature review

Farène G.K. Nouatin^a, Mori W. Gouroubera^{a,b,*}, Ismail Moumouni-Moussa^a^a Laboratory of Research on Innovation for Agricultural Development (LRIDA), University of Parakou, Benin^b International Center for Tropical Agriculture (CIAT), Dakar, Senegal

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ABSTRACT

Despite the recognized importance of agricultural extension approaches and farmers' innovation in addressing challenges within the agrifood sector, the existing literature on how these extension approaches influence farmers' innovativeness remains fragmented. This fragmentation provides limited guidance for policymakers and practitioners, hindering effective decision-making and the development of best practices. Following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, the review analyzed 33 empirical studies focused on enhancing farmers' innovativeness in Africa. This analysis was based on a search that yielded 2891 records published up to June 29, 2024. These records were meticulously screened to meet predefined inclusion criteria. The studies were sourced from the Web of Science and Scopus databases, with additional documents identified through Google Scholar. The findings reveal that (1) farmers' innovativeness varies across different extension approaches; (2) digital-based approaches promote the highest levels of innovation adoption, mainly for non-technological innovation, while integrated approaches perform best for technological innovations; (3) both digital-based and integrated approaches are more effective for knowledge acquisition; (4) digital-based and multi-stakeholder approaches are more effective for fostering continuous innovation; (5) capacity building, community and social networks, and access to resources and support emerged as the most significant drivers of farmers' innovativeness; and (6) positive impacts with a greater focus on environmental and economic dimensions were reported, although variability in outcomes suggests that context-specific factors play a crucial role. This review provides valuable insights for policymakers and practitioners aiming to improve agricultural extension services and foster farmers' innovativeness. It emphasizes the need for context-sensitive approaches that support farmers in navigating complex agricultural challenges.

1. Introduction

The complexity of agricultural development challenges requires innovative thinking for sustainable solutions (Klerkx, 2020; Maryono et al., 2024; Schut et al., 2014). A growing body of literature highlights that agricultural innovation is vitally important for addressing pressing global challenges such as food security, climate change, and economic development, particularly in developing

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* Corresponding author. Laboratory of Research on Innovation for Agricultural Development (LRIDA), University of Parakou, Benin.

E-mail address: gourouwm@gmail.com (M.W. Gouroubera).

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countries (Berrueta et al., 2021; Bhatta et al., 2017; Boulestreau et al., 2022; Brun et al., 2021; Colnago et al., 2021; Hang and Chen, 2021; Meynard et al., 2017). Innovation has long been equated with the development, transfer, and adoption of technology (Schut et al., 2014). This perspective overlooks farmers' innovation capacity (Hermans et al., 2021). Moreover, it is widely recognized that farmers' innovativeness, whether by adopting, adapting existing technologies, or developing new solutions, is important for sustainable agricultural practices (Alaie, 2020; Bakker et al., 2022; Dolinska and d'Aquino, 2016; Ogisi and Begho, 2023; Tambo and Wünscher, 2018). Scholars also recognize that innovations developed or adapted by farmers are often better suited to local conditions, are more cost-effective, and can spread rapidly within farming communities (Ensor and de Bruin, 2022; Hermans et al., 2021; Waters-Bayer et al., 2008). Therefore, fostering farmers' innovativeness to cope with complex agricultural development challenges is necessary (Dolinska and d'Aquino, 2016; Ensor and de Bruin, 2022).

Over time, various training or extension approaches have been developed to disseminate information with an increasing call to improve farmers' innovativeness (Klerkx, 2020). However, the effectiveness of these approaches in fostering farmers' innovativeness remains uncertain (Pawera et al., 2024). Modern agricultural systems are diverse and dynamic, requiring approaches that not only disseminate information but also actively engage farmers in problem-solving and innovation processes (Gachanja, 2023; Hermans et al., 2021; Maryono et al., 2024; Ogisi and Begho, 2023; Pawera et al., 2024; Tambo and Wünscher, 2018). Understanding how different training and extension approaches can best support farmers in their innovative activities is crucial for designing effective agricultural policies and programs that align with broader discussions on knowledge dissemination, learning processes, and the role of social interactions in innovation.

Various development organizations have implemented diverse approaches focused on transferring technologies from researchers to farmers, emphasizing adoption (Dolinska and d'Aquino, 2016; Hermans et al., 2021; Ogisi and Begho, 2023; Tambo and Wünscher, 2018). Early models of agricultural extensions, such as Rogers' diffusion of innovations theory (Rogers et al., 2014), categorize farmers based on their willingness to adopt new practices, from "laggards" to "innovators". These models have guided the development of extension strategies that promote the uptake of recommended practices through demonstrations, training, and the distribution of materials. Recent developments in agricultural extension have shifted toward more participatory and farmer-centric approaches (Klerkx, 2020). Extension approaches, particularly in Africa, like Farmer Field Schools, farmer-led research, ICT-based initiatives, collaborative learning, innovation platforms, and local adaptation of technologies, are encouraged for their active participatory approach (Gouroubera et al., 2025; Klerkx, 2020; Klerkx et al., 2012; Naika et al., 2021). These methods are praised for their potential to foster deeper engagement and innovation among farmers by building on their local knowledge and experiences.

Some related studies highlight the importance of adaptive, participatory, and ICT-integrated approaches to improving farmers' innovativeness. For instance, Bakker et al. (2022) illustrated how technical innovations co-created during Farmer Field Schools encouraged farmers to modify their practices at the farm level in Burkina Faso and Togo, West Africa. Similarly, Friis-Hansen and Duveskog (2012) reported that Farmer Field Schools in East Africa significantly increased productivity by adopting new practices. However, Barzola Iza et al. (2020) found that farmers' innovativeness in multi-stakeholder approaches, such as innovation platforms, is context-related, indicating that the same approach can lead to different outcomes. Moreover, Klerkx et al. (2012) emphasized the importance of innovation networks and collaboration among diverse stakeholders for effective knowledge sharing. Furthermore, Birner et al. (2009) reported that pluralistic systems with multiple service providers enhance the relevance and impact in developing countries. Additionally, recent studies by Davis et al. (2024), Gouroubera et al. (2023a,b,c), and Naika et al. (2021) highlighted the effectiveness of digital extension services in enhancing farmer decision-making.

Despite the growing interest in participatory approaches, there is still a lack of systematic evaluation regarding their impact on farmers' innovativeness. Empirical evidence remains fragmented, and it is unclear which specific approach is most effective under different conditions. Moreover, the interaction between various contextual factors (e.g., social, economic, and environmental) and training methods in influencing innovation remains underexplored.

Addressing these gaps is critical for several reasons. First, it can provide actionable insights into designing extension services that enhance farmers' innovativeness and are better aligned with farmers' needs and capacities (Klerkx, 2020). Second, understanding the contextual effectiveness of different approaches can help tailor interventions to specific agricultural settings, enhancing their impact (Dolinska and d'Aquino, 2016; Ensor and de Bruin, 2022; Klerkx, 2020; Pawera et al., 2024; Tambo and Wünscher, 2018; Waters-Bayer et al., 2008). Finally, this knowledge can contribute to theoretical frameworks on agricultural innovation, offering a more nuanced understanding of learning and knowledge exchange processes in diverse farming communities. Therefore, we explored diverse agricultural extension and training approaches and evaluated their impact on farmers' innovativeness by systematically reviewing the literature on the extension approach and innovation, drawing insights from 33 studies conducted in Africa.

The rest of the work is structured as follows: The next section introduces the analytical framework guiding data collection and analysis, which is grounded in the literature on innovation and its measurement (e.g., Ariza et al., 2013; Läßle et al., 2015; Salembier et al., 2021; Walder et al., 2019). Then follows the methodology section, adhering to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. The "Results" section highlights key findings on extension approaches and their effectiveness in fostering farmers' innovation, followed by a discussion, implications, and suggestions for future research.

2. Farmers' innovativeness assessment framework

Measuring innovation is a complex endeavor (Ariza et al., 2013; Fagerberg et al., 2005) because the concept encompasses a broad spectrum of activities, including the creation or adoption of innovations that can be new to the firm, new to the market, or new to the world (Läßle et al., 2015; OECD, 2013). Innovations are generally classified into four categories: product, process, market, and organizational innovations (Gault, 2013). Within the agricultural sector, there exists a vast array of new technologies, farm practices,

and management techniques that can be classified as agricultural innovations (Läpple et al., 2015). Notably, most innovations in agriculture are those new to the farm rather than new to the market or world (Läpple et al., 2015). Innovation in agriculture can also be categorized into two main types: technological and non-technological, both of which are pivotal at the farm level. Technological innovations encompass advancements that enhance agricultural practices through new tools and systems, such as precision agriculture and biotechnology, which improve productivity and resource efficiency (Feng, 2024; Yao et al., 2024). For instance, the implementation of IoT in farming allows real-time monitoring of agricultural processes, optimizing management. In contrast, non-technological innovations involve changes in organizational practices and business models that can significantly impact farm productivity and livelihoods. These innovations include cooperative farming, diversification strategies, and adaptive marketing, which help improve economic resilience among farmers (Brown and Roper, 2017; Fuentes and Soto, 2015). This context-dependent nature of innovation underscores its multifaceted character; thus, the same innovation may be perceived as either incremental or radical depending on the specific circumstances of different studies.

While there is no consensus on a specific measurement instrument for innovation, the literature provides various indicators that can be combined for a more robust assessment of the effectiveness of extension approaches in fostering farmers' innovativeness (e.g., Ariza et al., 2013; Läpple et al., 2015; Salembier et al., 2021; Walder et al., 2019). One commonly used indicator is research and development (R&D) expenditure on agriculture, which, while useful at the macro level, is often not applicable for measuring innovation at the farm level (Läpple et al., 2015). Alternative approaches to measuring innovativeness vary widely, ranging from binary or multinomial assessments to ordered or numeric measurements. For example, farmers' innovativeness can be assessed dichotomously through a simple yes/no question, a method documented extensively in the literature. For example, Diederer et al. (2003) analyzed the adoption of innovations by distinguishing between frontrunners (innovators and early adopters) and laggards (late and non-adopters), based on their position on the diffusion curve for the stated innovation, as evaluated by a team of experts. Ariza et al. (2013) proposed three methodologies for measuring farmers' innovativeness. The first methodology involves a structured catalog of technologies within the agricultural subsector, which includes information about the frequency of innovation and classifies innovations into major, intermediate, and minor categories based on expert evaluations. The second methodology introduces an innovation index that combines information about the technological level and frequency of observed innovations, assigning higher values to firms that implement less frequent or comparatively advanced innovations relative to their competitors. This method allows the usage of ordinary least squares (OLS) estimation procedures since the dependent variable is a non-negative index value ranging from 0 to 100. The third methodology builds upon the second to identify factors driving innovation processes in agricultural firms. Additionally, Läpple et al. (2015) developed an important innovation measurement index that views farmers' innovativeness as a function of the number, quality, or depth of implemented innovations, focusing on innovation adoption, knowledge acquisition, and continuous innovation.

Guided by the literature, we selected several relevant innovation indicators for our study, establishing metrics related to innovation adoption, knowledge acquisition, and continuous innovation (Fig. 1). This composite index aims to capture the complexity of agricultural innovation, allowing for a comparative analysis of innovation activities and their development across different extension approaches. For the *innovation adoption* indicators, we employ two distinct measures. The first uses a dichotomous approach (yes/no) to determine whether at least 50 % of farmers have adopted a given innovation, based on Rogers' classification of adopters, which includes innovators (2.5 %), early adopters (13.5 %), and the early majority (34 %). If the adoption rate is not specified, we consider the authors' affirmation that the majority has adopted the innovation.

Additionally, we assessed the speed of adoption in years to account for the time taken to adopt the innovation. The second indicator involves expert evaluations, where experts assign a weight to each innovation (product, process, market, or organization, technological vs non-technological) based on its level of innovativeness and implementation effort, rated on a scale from 1 to 10, with 10 being highly innovative. The *knowledge acquisition* indicator assesses the degree to which the extension approach has improved farmers' interactions with advisory services or other stakeholders and peers on non-scheme-related matters. This includes evaluating participation in learning and training as essential components of knowledge development and diffusion, as well as social learning, which is central to the innovation system (Knickel et al., 2009; Läpple et al., 2015). We use the same scale from 1 to 10 to assess

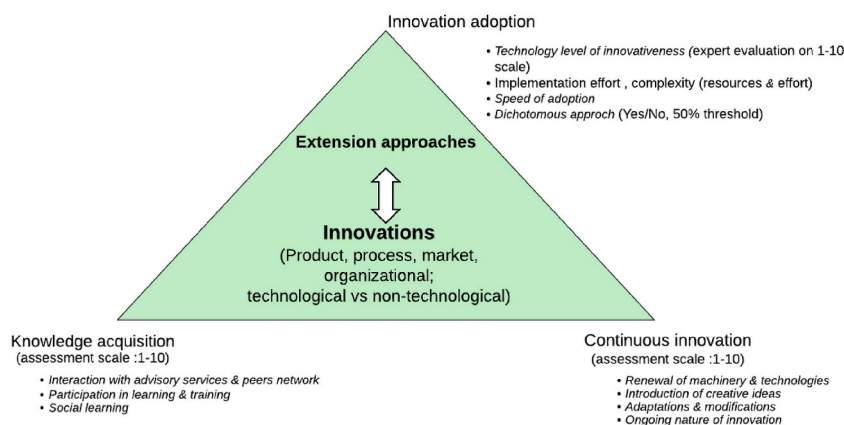


Fig. 1. Analytical framework.

knowledge acquisition. The *continuous innovation* indicator measures the extent to which an extension approach has led farmers to renew machinery, technologies, and introduce creative ideas, adaptations, or modifications during the study period, thereby evaluating the ongoing nature of innovation. Finally, contextual factors such as socio-economic status, environmental conditions, and institutional support were considered for their influence on the effectiveness of these approaches (Klerkx, 2012). This framework guides a systematic review of 33 papers in Africa, providing a detailed evaluation of how various training and extension methods impact farmers' innovativeness to identify successful strategies and contextual considerations that enhance agricultural innovation and facilitate the adoption of sustainable practices among farmers.

3. Methods

3.1. Systematic literature review

We carried out this systematic review following the PRISMA guidelines (Moher, 2009), ensuring that each step of the review process was transparent and thoroughly documented (Fig. 2). This review allowed us to identify various extension and training approaches and evaluate their impact on farmers' innovativeness in Africa.

3.2. Scope

We established clear inclusion and exclusion criteria tailored to the objectives of this study (Table 1). The systematic review encompasses studies focusing specifically on African farmers, investigating extension services, training programs, or educational initiatives to enhance their innovativeness. We included a diverse range of study designs, such as randomized controlled trials, case-control studies, cross-sectional studies, and qualitative studies, provided that they measured relevant outcomes, including the adoption of new technologies or practices and the development of innovative ideas. To ensure the relevance and accessibility of findings, we only considered studies published in English and French, with the full texts of the studies being accessible. Conversely, we excluded studies that did not focus on African farmers, did not address the specified interventions, lacked measurable innovation-related outcomes, or were opinion pieces or non-peer-reviewed articles. Additionally, studies published outside the defined scope, in languages other than French or English, or those whose full text was inaccessible were excluded from the review.

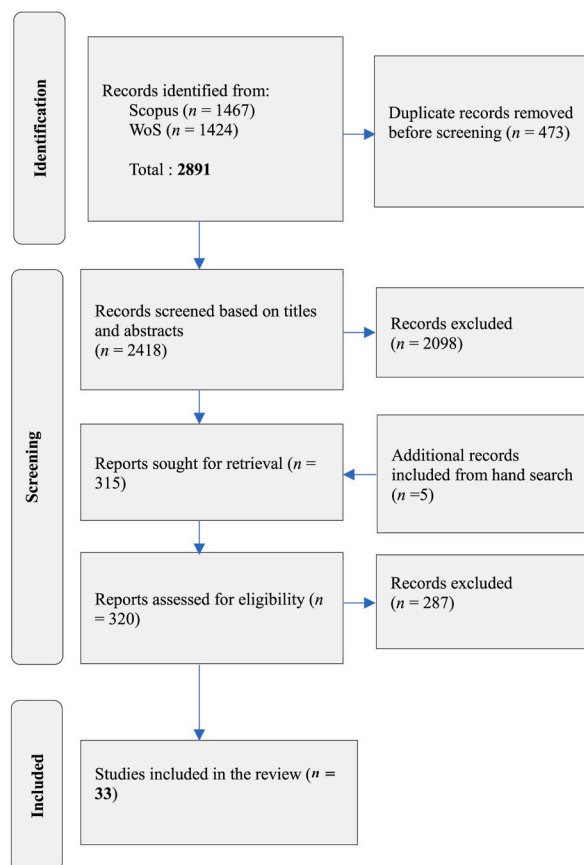


Fig. 2. Flow diagram (PRISMA).

Table 1
Eligibility criteria.

Criteria	Inclusion	Exclusion
Geographic Focus	Studies focusing on farmers in Africa.	Studies have not focused on African farmers.
Study Type	Empirical studies include randomized controlled trials, case-control studies, cross-sectional studies, and qualitative studies.	Opinion pieces, non-peer-reviewed articles, or studies lacking empirical data.
Language	Published in English or French.	Studies published in languages other than English or French.
Accessibility	Full text of peer-reviewed articles, books, and book chapters must be accessible.	Studies where the full text is inaccessible.
Intervention Focus	Studies examining extension services, training programs, or educational initiatives aimed at enhancing farmers' innovativeness.	Studies that do not address specified interventions related to extension or training.
Outcome Measurement	Studies measure relevant outcomes such as the adoption of new technologies, the development of innovative ideas, or other innovation-related outcomes.	Studies lack measurable innovation-related outcomes.

3.3. Search

Our search strategy utilized two main databases, i.e., Web of Science and Scopus. The search covered publications up to June 29, 2024, with no restrictions on publication date or language. We employed search strings that combined keywords related to training and extension approaches, innovation, and farmers as follows: (("training approach*" OR "extension approach*" OR "agricultural training" OR "extension service*" OR "Advisory service*" OR "extension program*" OR "capacity building" OR "knowledge transfer" OR "farmer education" OR "farmer training" OR "technology transfer" OR "innovation platform*" OR "farmer to farmer" OR "training and visit*" OR "farmer field school*" OR "phone-based" OR "mobile-based" OR "cell phone-based" OR "smartphone-based" OR "digital" OR "ICT") AND ("innovation" OR "innovative practice*" OR "farmer innovation" OR "agricultural innovation" OR "innovation adoption" OR "innovative capacity" OR "innovative behavior" OR "local innovation" OR "participatory innovation") AND ("farmer*" OR "smallholder farmer*" OR "agricultural producer*" OR "rural farmer*" OR "agricultural communit*")). The search yielded 2891 records (Scopus: $n = 1467$; Web of Science: $n = 1424$).

3.4. Screening

The screening process involved 2891 identified records from Scopus and Web of Science, based on the inclusion criteria (Fig. 2). The screening was performed using CADIMA, a systematic review screening software. The screening process began by removing 473 duplicates, resulting in 2418 unique records. The first stage involved title and abstract screening by two independent reviewers, who assessed the relevance of each study based on inclusion criteria, such as focusing on African farmers and examining training and extension approaches that influence farmers' innovativeness. A total of 2098 studies unrelated to the topic or lacking empirical data were excluded. For the records that passed this initial screening ($n = 315$), a full-text review was conducted to confirm their eligibility. Any discrepancies between reviewers' assessments were resolved through discussion or consultation with a third reviewer.

Additionally, a hand search of the references from the selected studies was performed to identify any further relevant articles, yielding five papers. The final process resulted in the exclusion of 287 papers. This thorough screening process resulted in a refined selection of 33 papers, providing a solid basis for analyzing the impact of training and extension approaches on innovation among African farmers (see Supplementary data-the list of included papers). The eligible studies underwent data extraction and analysis.

3.5. Data extraction and analysis

Data extraction for this systematic review involved a comprehensive and structured approach to gather all relevant information from the 33 included studies. A standardized data extraction form, implemented in Microsoft Excel, was used to capture all essential details. These details included study characteristics (such as authors, publication year, study design, and objectives), population information (including demographics, types of farmers, and geographical location), and specifics of the interventions (type of extension approach, description, duration, and delivery methods). Additionally, data on the types of innovations or technologies discussed in the studies, along with their levels of innovativeness, were collected. The form also included fields for outcome measures, such as quantitative metrics of innovation adoption, knowledge acquisition, and continuous innovation. Three independent reviewers from various backgrounds carried out the extraction process, assigning weights to each innovation, as well as to the innovativeness indicators (innovation adoption, knowledge acquisition, and continuous innovation), on a scale from 1 to 10, similar to the methods used by Ariza et al. (2013) and L  pple et al. (2015). These weighted scores from the three reviewers were then pooled to calculate the mean values, reflecting the specific innovativeness of each approach and technology. This process ensured a reliable and objective assessment of the innovations' impact across the studies.

4. Results

4.1. Overview of the included studies

The systematic review analyzed data from 33 studies conducted across 20 African countries, focusing on the effectiveness of

training and extension approaches in fostering farmer innovativeness. The studies addressed diverse objectives, from enhancing knowledge acquisition to promoting continuous innovation. A descriptive analysis illustrates the distribution of studies by year of publication (Fig. 3), indicating a growing recognition of the role of extension approaches in fostering farmers' innovativeness over time. Since 2009, the number of studies has varied from 1 to 4 per year, with over half (57.57 %; $n = 19$) published between 2018 and 2024. In terms of methodology, the majority of studies ($n = 18$) employed mixed methods (54.54 %) (e.g., Dayamba et al., 2018; Dosso et al., 2023; Karubanga et al., 2017; Kiptot and Franzel, 2015; Osumba et al., 2021; Zoundji et al., 2024), while 27.27 % were qualitative ($n = 9$) (e.g., Bakker et al., 2021; Gunnarsson, 2021; Hermans et al., 2021; Lema et al., 2021) and 18.18 % were quantitative ($n = 6$) (e.g., Faturoti et al., 2008; Lawal et al., 2019; Nakano et al., 2018; Okori et al., 2022; Tambo et al., 2019; Wellard et al., 2013). Qualitative studies have focused primarily on descriptive analyses of cases, outlining the outcomes of training and extension approaches in fostering farmer innovativeness. Some quantitative and mixed-method studies have utilized experimental or quasi-experimental designs to estimate the effectiveness of these extension approaches (e.g., Nakano et al., 2018; Tambo et al., 2019; Zoundji et al., 2024).

Regarding geographical representation, Uganda has the highest representation, with 13.04 % ($n = 8$) of the studies, highlighting a significant focus on the impacts of training and extension services in this region. Benin, Ghana, and Tanzania each account for 8.70 % ($n = 4$), reflecting strong interest in the effectiveness of training and extension approaches in these countries. Ethiopia, Kenya, and Malawi (each at 6.52 %, $n = 3$) emphasize the importance of East African agricultural systems, which face unique challenges requiring effective strategies. Countries such as Mali, Morocco, Mozambique, Nigeria, and Togo (each at 4.35 %, $n = 2$) provide valuable insights into the broader implications of training and extension services. Burkina Faso, Cameroon, Gambia, Zambia, Senegal, Tunisia, and Zimbabwe (each at 2.17 %, $n = 1$) indicate emerging areas of interest that may warrant further exploration.

4.2. Sociodemographic characteristics of the farmers included in the studies

Most of the studies (75.75 %, $n = 25$) involved farmers who were part of associations, indicating a strong inclination toward collective action, which can enhance knowledge sharing and innovation. The average age of the farmers included in the review is 42.7 years ($n = 13$), with an average of 14.39 years of agricultural experience ($n = 6$). This experience is beneficial as it suggests familiarity with local agricultural practices and challenges. The average farm size reported is 3 ha ($n = 5$). This size is typical for smallholder farmers in many African countries, indicating that most farmers operate on limited land. Additionally, the average household size among the farmers is 6.51 members ($n = 7$) (Table 2). Larger household sizes may indicate a higher demand for food production and resources, motivating farmers to innovate and improve their agricultural practices.

4.3. Types of training and extension approaches

The included studies explored various training and extension approaches, which can be categorized based on their delivery methods and facilitators (Table 3). Participatory approaches, such as Farmer Field Schools and farmer-to-farmer extension, were the most common, each utilized in 18.18 % ($n = 6$) of the studies (Afranaa Kwapong and Ankrah, 2023; Bakker et al., 2021, 2022; Fa, 2018; Hermans et al., 2021; Osumba et al., 2021). Traditional extension services, like Training and Visits, were employed in 6.06 % ($n = 2$) of the studies (Asare-Nuamah et al., 2019; Faure et al., 2013). Digital-based approaches, including farmer-to-farmer videos (12.12 %, $n = 4$) (Karubanga et al., 2017; Zossou et al., 2009, 2010; Zoundji et al., 2024), smartphone-based training (6.06 %, $n = 2$) (Daniso et al., 2020), and a combination of SMS, radio, and video (3.03 %, $n = 1$) (Dolinska et al., 2020; Tambo and Wünscher, 2018), were also utilized to enhance the reach and effectiveness of extension services. Integrated approaches, which combined multiple methods (Dayamba et al., 2018; Dolinska et al., 2020; Dolinska and d'Aquino, 2016; Gunnarsson, 2021; Karubanga et al., 2016; Okori et al., 2022), such as participatory integrated approaches, farmer-to-farmer extension with Farmer Field Schools, video-mediated and face-to-face extension, and community-based approaches with training and farmer-to-farmer extension, were used in 18.18 % ($n = 6$) of the studies. Multi-stakeholder approaches, particularly innovation platforms, were employed in 15.15 % ($n = 5$) of the studies to foster collaborative learning and problem-solving (Lawal et al., 2019; Lema et al., 2021; Parry et al., 2020; Spurk et al., 2023; Tur-yahikayo et al., 2019). The training or extension services providers (Fig. 4) ranged from the government (37 %, $n = 12$) (e.g., Afranaa Kwapong and Ankrah, 2023; Lema et al., 2021; Munthali et al., 2018), private sector (12 %, $n = 4$) (e.g., Dolinska and d'Aquino, 2016; Gunnarsson, 2021), NGOs (27 %, $n = 9$) (e.g., Dosso et al., 2023; Osumba et al., 2021; Wellard et al., 2013), farmer associations (15 %, $n = 5$) (e.g., Daniso et al., 2020; Faysse et al., 2012; Okori et al., 2022) and research institutions (9 %, $n = 3$) (e.g., Dayamba



Fig. 3. Distribution of studies by publication year.

Table 2
Sociodemographic characteristics.

	Mean	Number of studies
Age	42.7	13
Farm size	3	5
Farmers' experience	14.39	6
Household size	6.51	7

Table 3
Distribution of extension approaches.

	Extension approaches	% (frequency)	General
Participatory approaches	Farmer Field School	18.18 (<i>n</i> = 6)	36.36 (<i>n</i> = 12)
	Farmer to farmer	18.18 (<i>n</i> = 6)	
Traditional extension approaches	Training and Visits	6.06 (<i>n</i> = 2)	6.06 (<i>n</i> = 2)
Digital-based approaches	Farmer-to-farmer video	12.12 (<i>n</i> = 4)	21.21 (<i>n</i> = 7)
	Smartphone	6.06 (<i>n</i> = 2)	
Integrated approaches	SMS/Radio/Video	3.03 (<i>n</i> = 1)	18.18 (<i>n</i> = 6)
	Participatory integrated approach	3.03 (<i>n</i> = 1)	
	Farmer-to-farmer and Farmer Field Schools	3.03 (<i>n</i> = 1)	
	Video-mediated and face-to-face extension	3.03 (<i>n</i> = 1)	
	Training and Visit, Farmer to Farmer, and Agricultural subsidies	3.03 (<i>n</i> = 1)	
	Learning networks and farmer field schools.	3.03 (<i>n</i> = 1)	
	Community-based extension approaches and Training. Farmer to farmer	3.03 (<i>n</i> = 1)	
Multi-stakeholder approaches	Innovation platforms	15.15 (<i>n</i> = 5)	18.18 (<i>n</i> = 6)
	Total	100 (<i>n</i> = 33)	

et al., 2018; Faturoti et al., 2008; Razafinimpiasa et al., 2022). Most studies have also addressed gender sensitivity in their training approaches (76 %, *n* = 25), highlighting the inclusion of women and marginalized groups.

4.4. Innovations and technologies promoted

The studies included in the review identified a broad spectrum of innovations and technologies aimed at enhancing agricultural practices across Africa, categorized into various types that reflect the diverse needs and challenges faced by farmers (Fig. 5). The innovations can be classified into process innovations (53 %, *n* = 25) (e.g., Lawal et al., 2019; Spurk et al., 2023; Zossou et al., 2010), product innovations (36 %, *n* = 17) (e.g., Dayamba et al., 2018; Faysse et al., 2012; Okori et al., 2022; Tambo and Wünsch, 2018), organizational innovations (7 %, *n* = 3) (e.g., Daniso et al., 2020; Dolinska and d'Aquino, 2016; Fa, 2018), and market innovations (4 %, *n* = 2) (e.g., Osumba et al., 2021). The different types of innovations were related to agricultural development and sustainability (Table 4). Most of the innovations were related to crop improvement (*n* = 13), such as improved maize and sorghum varieties, modern rice varieties, and plant hybrid technology, which are designed to boost yields and enhance resilience against pests and diseases (e.g., Asare-Nuamah et al., 2019; Okori et al., 2022; Turyahikayo et al., 2019). Sustainable practices (*n* = 11) were also covered by the studies, including conservation agriculture and agroecological vegetable cultivation practices (e.g., Dosso et al., 2023; Faure et al., 2013; Zoundji et al., 2024). Other important innovations in soil fertility management (*n* = 11) have emphasized proper fertilizer application techniques, improved fertilizer provision, and the use of soil monitoring tools, which are essential for optimizing crop production and maintaining soil health (e.g., Bakker et al., 2022; Razafinimpiasa et al., 2022; Spurk et al., 2023; Tambo et al., 2019). Furthermore, climate information services (*n* = 6), such as seasonal climate forecasts, equip farmers with the necessary tools to adapt to climate variability and mitigate its impacts (e.g., Dolinska et al., 2020; Hermans et al., 2023; Lawal et al., 2019; Munthali et al., 2018). In livestock management (*n* = 4), innovations focused on improving feeding systems and included a dairy farming simulation

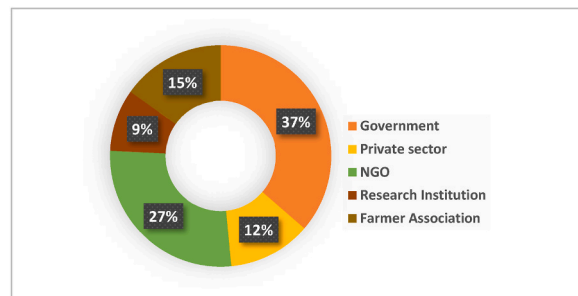


Fig. 4. Extension services provider.

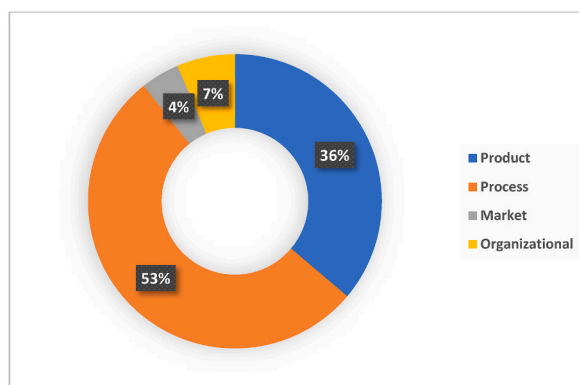


Fig. 5. Type of innovation.

Table 4
Innovations.

Category	Innovation	Technological	n
Crop improvement (n = 13)	Improved maize varieties	Yes	2
	Improved sorghum varieties	Yes	1
	Improved crop seed	Yes	2
	New crops (sugar cane, sugar beet, and paddy)	Yes	1
	Plantain hybrid technology	Yes	2
	Rice modern varieties	Yes	3
	Intercropping	Yes	2
Livestock management (n = 4)	Livestock feeds and feeding systems	Yes	3
	Dairy farming (simulation game)	Yes	1
Sustainable practices (n = 11)	Conservation no-tillage agriculture	Yes	1
	Conservation agriculture practices	Yes	5
	Agroecological vegetable cultivation practices	Yes	1
Pest and disease management (n = 5)	Sustainable agricultural techniques and land management	Yes	4
	Detection and control of pests (Fall Armyworms)	Yes	2
	Aflatoxin mitigation methods	Yes	1
	Information on pest occurrences	No	1
	Biopesticides	Yes	1
Soil fertility management (n = 11)	Proper application of fertilizers	Yes	4
	Phosphate fertilizer usage	Yes	1
	Improved fertilizer provision	Yes	3
	Soil monitoring tools (Chameleon, Wetting Front Detectors)	Yes	1
	Composting	Yes	2
	Climate-smart services	No	1
Climate-smart agriculture (n = 6)	Micro-dosing innovations	Yes	1
	Improved irrigation techniques	Yes	2
	Seasonal climate forecasts	No	1
	Information on drought occurrences	No	1
	Contract farming	No	1
Market access (n = 2)	Common bean input and output trading	No	1
	Mobile phone usage for rural innovation services	No	1
Digital and improved techniques (n = 5)	Participatory decision-making tools	No	2
	Improved rice parboiling technology	Yes	2

game that enhances farmers' understanding of best practices through interactive learning (e.g., Kiptot and Franzel, 2015; Lema et al., 2021; Wellard et al., 2013). Pest and disease management innovations ($n = 5$), such as methods for detecting and controlling Fall Armyworms, aflatoxin mitigation strategies, and the use of biopesticides, are critical in addressing the challenges posed by pests and diseases that threaten crop yields and food security (e.g., Munthali et al., 2018; Okori et al., 2022; Turyahikayo et al., 2019). Market access innovations ($n = 2$), including contract farming and common bean input-output trading, aim to enhance farmers' market positions and ensure fair pricing for their produce (Osumba et al., 2021). Finally, digital and improved techniques ($n = 5$), such as mobile phone usage for rural innovation services and participatory decision-making tools, facilitate better information dissemination and farmer engagement, which are also promoted, ultimately fostering a more inclusive agricultural ecosystem (e.g., Daniso et al., 2020; Fa, 2018; Zossou et al., 2010). In terms of technological classification, the results show that 75 % of the innovations are technological, whereas 25 % are non-technological.

4.5. Influence of extension approaches on farmers' innovativeness

4.5.1. Innovation adoption and speed

Most studies (90.90 %, $n = 30$) indicate that the extension approaches implemented facilitated the adoption of innovation by over 50 % of farmers. The speed of innovation adoption varied across different extension approaches (Table 5). Innovation platforms showed the quickest adoption time of 2 years, albeit with limited data ($n = 1$). Integrated approaches closely with a mean of 2.75 years. Digital-based approaches adoption time was 3 years. Participatory approaches, particularly Farmer Field Schools, had a longer mean adoption time of 4.17 years. These findings highlight the varying effectiveness and efficiency of extension approaches in promoting innovation adoption among farmers, suggesting that tailored approaches may be necessary to optimize outcomes based on specific contexts and farmer needs.

4.5.2. Technology level of innovativeness

The technology level of innovativeness measures the extent to which farmers have adopted innovations or technologies on a scale from 1 to 10, with 10 being highly innovative. The results (Table 6) show that digital-based approaches had the highest mean score of 7.57 (SD = 0.787), indicating that these approaches were most effective in fostering the adoption of innovative technologies among farmers. Integrated approaches, multi-stakeholder approaches, and participatory approaches had similar mean scores (6.33, 6.17, and 6.33, respectively), suggesting that they were also effective in promoting technological innovation, albeit to a lesser degree than digital-based approaches. Traditional extension services had the lowest mean score of 5.00 (SD = 1.414), implying that they were the least effective in fostering technological innovation among farmers.

The comparison of adoption potential between technological and non-technological innovations across various extension approaches reveals distinct patterns (Table 7). Non-technological innovations, such as climate-smart services and mobile phone usage for rural innovation services tend to achieve higher adoption rates through digital-based (score: 9.0), participatory (9.5), and integrated (8.5) approaches. These methods emphasize accessibility, local relevance, and two-way communication, making them well-suited for disseminating information-driven innovations. In contrast, technological innovations, such as improved seeds, irrigation systems, or processing equipment, show the highest adoption scores under integrated approaches (8.0), followed by multi-stakeholder approaches (7.5), where access to inputs, technical training, and institutional coordination is better facilitated. Digital-based and participatory approaches also support technological adoption but to a lesser extent, with mean scores of 6.5 and 6.0, respectively, which are often limited by the need for physical resources and infrastructure. Traditional extension services scored lowest for both types (5.5 for technological innovation and 6.5 for non-technological innovation), highlighting their limited effectiveness in scaling modern innovations.

4.5.3. Knowledge acquisition

The findings reveal (Table 6) that digital-based approaches scored the highest in knowledge acquisition, with a mean of 7.57 (SD = 1.272), indicating that these approaches significantly improved farmers' access to information and resources. The interactive nature of digital platforms, such as mobile applications and farmer-to-farmer video, facilitates real-time communication and knowledge sharing among farmers, extension agents, and agricultural experts, enhancing the overall learning experience. The integrated approaches followed closely with a mean score of 7.17 (SD = 1.602), suggesting that these methods effectively combine various learning modalities to promote knowledge exchange and collaboration. Multi-stakeholder approaches, with a mean score of 6.00 (SD = 0.894), also contributed to knowledge acquisition by fostering partnerships between farmers, researchers, and extension agents. However, their impact was slightly less pronounced than digital and integrated approaches. Participatory approaches, which emphasize farmer engagement and peer learning, achieved a mean score of 6.75 (SD = 1.215). These approaches encourage farmers to share their experiences and insights, facilitating a collaborative learning environment that enhances knowledge retention and application. Traditional extension services, however, had the lowest mean score of 5.00 (SD = 1.414), indicating that they were less effective in promoting knowledge acquisition compared to more innovative approaches. This suggests that conventional extension approaches may not fully engage farmers or address their diverse learning needs, limiting their potential for knowledge development.

4.5.4. Continuous innovation

The continuous innovation indicator assesses the extent to which extension approaches have encouraged farmers to renew machinery and tools, adopt new technologies, and introduce creative ideas, adaptations, or modifications throughout the experimentation period. This ongoing nature of innovation is crucial for enhancing agricultural productivity and sustainability, as it reflects farmers' ability to respond to changing environmental conditions and market demands. The findings indicate that digital-based approaches scored the highest in promoting continuous innovation (Table 6), with a mean of 6.43 (SD = 1.718). This suggests that these

Table 5
Speed of technology adoption.

Extension approaches	<i>n</i>	Mean (years)	SD
Digital-based approaches	3	3	3.46
Integrated approaches	4	2.75	1.50
Innovation platform	1	2	–
Participatory approaches	6	4.17	1.72

Table 6
Farmers' innovativeness.

Innovation indicators	Extension approaches	n	Mean	SD
Technology level of innovativeness (innovation adoption)	Digital-based approaches	7	7.57	0.787
	Integrated approaches	6	6.33	1.751
	Multi-stakeholder approaches	6	6.17	1.941
	Participatory approaches	12	6.33	1.497
	Traditional extension services	2	5.00	1.414
Knowledge acquisition	Digital-based approaches	7	7.57	1.272
	Integrated approaches	6	7.17	1.602
	Multi-stakeholder approaches	6	6.00	0.894
	Participatory approaches	12	6.75	1.215
	Traditional extension services	2	5.00	1.414
Continuous innovation	Digital-based approaches	7	6.43	1.718
	Integrated approaches	6	4.17	3.488
	Multi-stakeholder approaches	6	5.67	1.033
	Participatory approaches	12	3.83	2.443
	Traditional extension services	2	2.50	2.121

Table 7
Comparative analysis of innovation adoption across extension approaches.

Extension Approach	Technological innovation (SD)	Non-technological innovation (SD)
Digital-based approaches	6.5 (1.2)	9.0 (0.8)
Integrated approaches	8.0 (0.7)	8.5 (0.6)
Multi-stakeholder approaches	7.5 (1.0)	8.0 (0.9)
Participatory approaches	6.0 (1.3)	9.5 (0.5)
Traditional extension services	5.5 (1.5)	6.5 (1.2)

approaches effectively facilitate access to information and resources, enabling farmers to experiment with and adopt new technologies and practices. For example, the study of [Zossou et al. \(2009\)](#) conducted in Benin showed how video can trigger innovation. The interactive nature of digital platforms allows real-time feedback and support, fostering an environment where farmers feel empowered to innovate continuously.

In contrast, integrated approaches had a mean score of 4.17 (SD = 3.488), indicating a moderate level of continuous innovation. While these approaches combine various methods to enhance learning and collaboration, the variability in their effectiveness suggests that they may not always lead to sustained innovation. Multi-stakeholder approaches also showed promise, with a mean score of 5.67 (SD = 1.033), highlighting their potential to foster collaborative innovation among diverse actors in the agricultural sector. However, participatory approaches had a lower mean score of 3.83 (SD = 2.443), suggesting that while they promote engagement and knowledge sharing, they may not consistently lead to ongoing technological adoption or creative adaptations. Traditional extension services, with the lowest mean score of 2.50 (SD = 2.121), were the least effective in fostering continuous innovation among farmers. This indicates that conventional methods may lack the dynamism and responsiveness required to encourage farmers to actively embrace new ideas and practices. The findings emphasize the need for extension approaches that not only provide initial training but also support sustained engagement and learning over time.

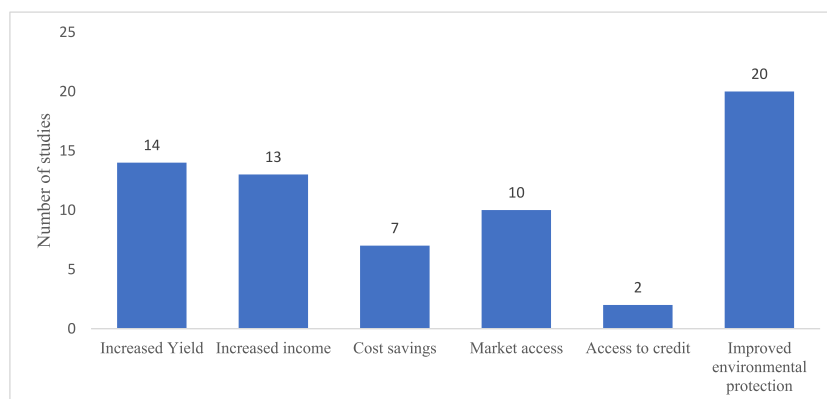


Fig. 6. Impacts of interventions on farmers.

4.6. Impact on smallholder farmers

The review analyzes various impacts of innovative activities on smallholder livelihoods, focusing specifically on the effects directly linked to intervention activities. We present the impacts of innovations as documented in the included studies, synthesizing critical findings from quantitative and qualitative impact evaluations. Among the 33 documents analyzed, 28 examined the impact of the interventions on at least one aspect of the economic or environmental dimensions of smallholder livelihoods. Few studies employed a quantitative approach ($n = 5$), while 23 relied on qualitative methods. This gap highlights the need for more rigorous research designs to effectively assess the impacts of innovations on farmers' livelihoods. Fig. 6 illustrates that the most discussed dimension was environmental benefits, highlighted in 20 studies, indicating a growing recognition of the importance of sustainable practices in enhancing resilience and ensuring long-term productivity. The studies also reported various economic impact indicators, including increased yield ($n = 14$), increased income ($n = 13$), market access ($n = 10$), and credit access ($n = 2$).

4.6.1. Impact on agricultural productivity

The review identified a significant positive impact of innovative activities on agricultural productivity, with an increase in yield being the most commonly reported outcome ($n = 14$) (Afranaa Kwapong and Ankrah, 2023; Bakker et al., 2021; Dosso et al., 2023; Faturoti et al., 2008; Gunnarsson, 2021; Kiptot and Franzel, 2015; Lawal et al., 2019; Nakano et al., 2018; Okori et al., 2022; Parry et al., 2020; Razafinimpiasa et al., 2022; Spurk et al., 2023; Turyahikayo et al., 2019; Wellard et al., 2013). A majority of the studies that reported yield increases employed a qualitative approach. For instance, Bakker et al. (2021) and Lawal et al. (2019) utilized qualitative approaches and noted significant yield improvements associated with Farmer Field Schools and innovation platforms, respectively. While qualitative data predominated, some studies provided quantitative evidence of yield enhancements. Notably, a remarkable yield improvement of up to 300 % in maize production was reported in Ghana, which was attributed to the Farmer Field School approach (Afranaa Kwapong and Ankrah, 2023). Similarly, Okori et al. (2022) documented a 1.83 % increase among groundnut farmers in Malawi, while Wellard et al. (2013) reported a 50 % yield increase among farmers in Ghana, Uganda, and Malawi through a combination of community-based extension approaches, training, and farmer-to-farmer exchanges. Additionally, Nakano et al. (2018) observed a 2.2 % increase in rice yields among farmers in Tanzania. OKTuryahikayo et al. (2019) reported a yield increase of up to 25 % in Uganda using the Farmer Field Schools approach.

4.6.2. Impact on income

A total of 13 documents reported positive impact of the interventions on farmers' income (Afranaa Kwapong and Ankrah, 2023; Daniso et al., 2020; Dosso et al., 2023; Faturoti et al., 2008; Faysse et al., 2012; Kiptot and Franzel, 2015; Lawal et al., 2019; Nakano et al., 2018; Okori et al., 2022; Razafinimpiasa et al., 2022; Spurk et al., 2023; Turyahikayo et al., 2019; Zossou et al., 2010). For example, in Ghana, Afranaa Kwapong and Ankrah (2023) reported a positive increase in the income of maize farmers. Similarly, Razafinimpiasa et al. (2022) documented increased income in Madagascar, indicating the effectiveness of using improved fertilizers in enhancing rice farmers' financial well-being. In Malawi, a notable increase of 77 % was reported by Okori et al. (2022), demonstrating significant economic benefits from using community seed banks and the farmer-to-farmer extension approach. Nakano et al. (2018) found a modest increase of 0.84 % in rice farmers' income with the farmer-to-farmer approach in Tanzania, while Turyahikayo et al. (2019) reported a 20 % increase in the income of farmer members of the innovation platform in Uganda, showing varying degrees of success across different regions. Notably, the presence of multiple studies indicating increased income suggests a trend toward positive economic impacts resulting from innovative agricultural practices. However, it is important to recognize that not all studies reported increased income. For instance, several studies conducted in countries such as Kenya, Benin, Uganda, and Morocco did not find significant income improvements (e.g., Bakker et al., 2021; Fa, 2018; Karubanga et al., 2016; Lema et al., 2021; Osumba et al., 2021; Wellard et al., 2013; Zoundji et al., 2024), highlighting the variability in outcomes based on context, intervention type, and local conditions. These mixed results indicate the need for further research to identify the factors contributing to successful income increases in different agricultural settings.

4.6.3. Impact on cost savings

The analysis of cost savings among farmers reveals important insights into the economic impacts of agricultural interventions across various countries. While many studies do not report cost savings, several notable exceptions indicate that specific practices ($n = 7$) can lead to significant reductions in expenses (Dosso et al., 2023; Kiptot and Franzel, 2015; Lawal et al., 2019; Parry et al., 2020; Spurk et al., 2023; Turyahikayo et al., 2019; Wellard et al., 2013). Most of these studies used a qualitative approach ($n = 6$). For instance, Wellard et al. (2013) documented farmers' cost savings in Ghana, Uganda, and Malawi, suggesting that sustainable agriculture technologies and improved livestock management can effectively lower operational costs for farmers. In Benin, Dosso et al. (2023) also found evidence of cost savings, indicating that the combination of extension approaches strengthening farmers' innovativeness in sustainable land management can alleviate financial burdens on farmers. Turyahikayo et al. (2019), using a quantitative approach, reported cost savings of 15 % among innovation platform members in Uganda, demonstrating tangible economic benefits from the implemented practices. Additionally, studies by Spurk et al. (2023) and Lawal et al. (2019) in Nigeria confirmed that cost savings were achieved, reinforcing the notion that innovative agricultural practices can lead to reduced expenditures.

4.6.4. Impact on market access

A total of 10 studies reported improvements in farmers' market access (Faturoti et al., 2008; Gunnarsson, 2021; Kiptot and Franzel, 2015; Lema et al., 2021; Munthali et al., 2018; Okori et al., 2022; Tambo et al., 2019; Turyahikayo et al., 2019; Zossou et al., 2010;

Zoundji et al., 2024). For instance, in Benin, Zoundji et al. (2024) reported positive market access to agroecological vegetable farmers after watching training videos for supporting agroecological vegetable production, indicating that the interventions have successfully improved farmers' ability to reach markets and sell their products. Similar results were found in other regions, such as in Malawi, with the combination of farmer-to-farmer and farmer-field schools (Okori et al., 2022), in Uganda, using a digital-based approach (radio, SMS, and video) (Tambo et al., 2019), and in Gambia with farmers' field schools (Gunnarsson, 2021).

4.6.5. Impact on access to credit

Out of the 33 documents analyzed, only two (Daniso et al., 2020; Kiptot and Franzel, 2015) indicated positive access to credit, although it is an important factor. This shows potential barriers faced in obtaining necessary financing for agricultural activities. For example, research by Afranaa Kwapong and Ankrah (2023) in Ghana, as well as studies by Asare-Nuamah et al. (2019) and Osumba et al. (2021), did not find evidence of improved access to credit, suggesting that farmers in these regions may struggle to secure loans or financial assistance.

4.6.6. Environmental impact

The analysis of the environmental impacts of various agricultural interventions reveals a significant trend toward positive outcomes across multiple studies and countries. A total of 20 documents reported positive farmer interactions with the environment, such as soil health ($n = 7$) (Afranaa Kwapong and Ankrah, 2023; Bakker et al., 2021; Dosso et al., 2023; Faure et al., 2013; Lawal et al., 2019; Spurk et al., 2023; Wellard et al., 2013), water management ($n = 3$) (Dayamba et al., 2018; Fa, 2018; Parry et al., 2020), biodiversity and crop resilience ($n = 6$) (Bakker et al., 2022; Okori et al., 2022; Osumba et al., 2021; Razafinimpiasa et al., 2022; Tambo et al., 2019; Turyahikayo et al., 2019), and climate adaptation and resilience ($n = 4$) (Hermans et al., 2021; Kiptot and Franzel, 2015; Zossou et al., 2009; Zoundji et al., 2024). The various positive impacts revealed in these studies underscore the importance of these interventions in promoting sustainability.

4.7. Drivers of farmers' innovativeness

The review identified several key drivers of farmers' innovativeness (Fig. 7), reflecting the multifaceted nature of agricultural innovation. Capacity building ($n = 15$) emerged as the most significant driver, emphasizing the importance of enhancing farmers' skills and knowledge through targeted training programs (e.g., Asare-Nuamah et al., 2019; Dosso et al., 2023; Fayssse et al., 2012; Hermans et al., 2023; Kiptot and Franzel, 2015; Lawal et al., 2019; Lema et al., 2021; Parry et al., 2020; Turyahikayo et al., 2019). Community and social networks ($n = 14$) also play crucial roles, as farmers who engage with their peers are more likely to share knowledge and collaborate on innovative practices (Bakker et al., 2021, 2022; Dayamba et al., 2018; Dolinska and d'Aquino, 2016; Fa, 2018; Nakano et al., 2018; Okori et al., 2022; Parry et al., 2020; Turyahikayo et al., 2019; Zossou et al., 2010; Zoundji et al., 2024). Access to resources and support ($n = 11$) is essential for enabling farmers to experiment with new technologies (e.g., Gunnarsson, 2021; Kiptot and Franzel, 2015; Lema et al., 2021; Munthali et al., 2018; Tambo and Wünsch, 2018; Tambo et al., 2019; Turyahikayo et al., 2019; Zossou et al., 2009, 2010). Additionally, the influence of technology and innovations ($n = 8$) was also indicated as drivers of farmers' innovativeness (e.g., Dayamba et al., 2018; Hermans et al., 2023; Karubanga et al., 2017; Munthali et al., 2018; Spurk et al., 2023; Zossou et al., 2010), highlighting the necessity of integrating modern agricultural practices to improve productivity. For example, Dolinska and d'Aquino (2016) indicate that simulation games to test innovations and encourage experimentation among farmers further stimulate farmers' innovativeness in Tunisia, Mozambique, and Ethiopia. The same observation was made with farmer-to-farmer videos in Benin (Zossou et al., 2009, 2010; Zoundji et al., 2024) and Uganda (Karubanga et al., 2016, 2017). Other

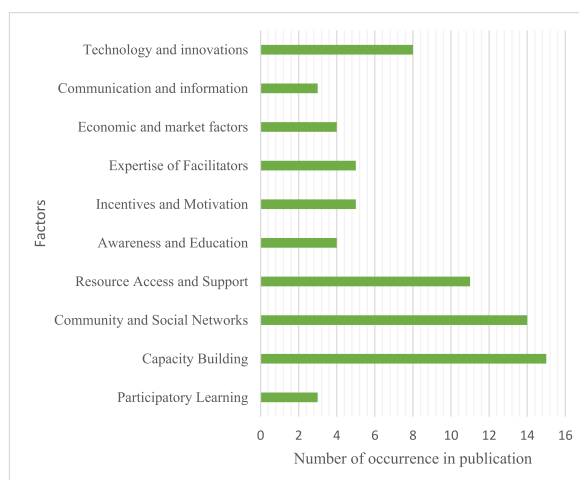


Fig. 7. Drivers of farmers' innovativeness.

notable drivers include incentives and motivation ($n = 5$) (Dosso et al., 2023; Fa, 2018; Kiptot and Franzel, 2015; Munthali et al., 2018), which encourage farmers to adopt new practices, and the expertise of facilitators ($n = 5$) (Fa, 2018; Lema et al., 2021; Munthali et al., 2018; Tambo et al., 2019; Turyahikayo et al., 2019), who facilitate knowledge transfer. Economic and market factors ($n = 4$) (Afranaa Kwapong and Ankrah, 2023; Lawal et al., 2019; Okori et al., 2022; Zoundji et al., 2024) and effective communication and information dissemination ($n = 3$) (Lawal et al., 2019; Spurk et al., 2023; Tambo et al., 2019) further support farmers in their innovative efforts. Together, these drivers create an environment conducive to fostering innovation among farmers.

5. Discussion

The findings of this systematic literature review underscore the critical role that agricultural extension approaches play in fostering farmers' innovativeness across diverse contexts in Africa. The review analyzed 33 studies, revealing significant variations in innovation outcomes associated with different extension approaches. This section discusses the implications of these findings, the contextual factors influencing effectiveness, and recommendations for future research and practice.

5.1. Variability of innovation outcomes

The review highlights that not all agricultural extension approaches yield the same results in enhancing farmers' innovativeness. This variability highlights the importance of understanding the specific contexts and approaches to successful innovation outcomes.

5.1.1. Digital-based approaches enhance farmers' innovativeness

One of the key findings of this review is the effectiveness of digital tools in promoting innovation adoption, knowledge acquisition, and continuous innovation among farmers. Digital-based extension services, such as mobile applications, farmer-to-farmer video usage, and online platforms, significantly enhance farmers' access to information and resources, ultimately leading to higher rates of innovation adoption (Davis et al., 2024; Gouroubera et al., 2023a,b,c). These digital tools can provide farmers with real-time information on weather, market prices, and best practices, empowering them to make more informed decisions about adopting innovative technologies and practices (Naika et al., 2021). Moreover, digital extension services can overcome some of the limitations of traditional extension approaches, such as limited reach and scalability. By leveraging mobile technologies, these services can provide personalized advice and support to farmers, even in remote areas (Klerkx, 2020; Klerkx et al., 2012). Additionally, digital platforms can facilitate knowledge sharing and collaboration among farmers, extension workers, and researchers, creating a more connected and responsive agricultural innovation system (Klerkx et al., 2012). Additionally, several studies have shown that farmer-to-farmer videos tailored to their context—in local languages—are effective in triggering farmers' innovativeness (Bentley et al., 2014; Gouroubera et al., 2023a,b,c; Zossou et al., 2009, 2010; Zoundji et al., 2024). However, the effectiveness of digital tools in enhancing farmers' innovativeness is not without challenges. Issues such as limited digital literacy, lack of infrastructure, and affordability can hinder the adoption and use of these technologies, particularly among marginalized farming communities (Ensor and de Bruin, 2022). Moreover, the integration of digital tools into existing extension systems requires significant investments in training, capacity building, and institutional support (Gouroubera et al., 2023a,b,c; Klerkx, 2020, 2021). Despite these challenges, the potential of digital tools to enhance farmers' innovativeness is significant. As innovation continues to advance and become more accessible, digital extension services will likely play an increasingly important role in promoting sustainable agricultural practices and empowering farmers to innovate (Davis et al., 2024).

5.1.2. Effectiveness of participatory approaches

Another key finding is the effectiveness of participatory extension approaches in promoting farmers' innovativeness. Approaches like Farmer Field Schools and farmer-to-farmer initiatives have been particularly effective in engaging farmers and facilitating the adoption of innovative practices. These approaches actively involve farmers in the learning process, allowing them to experiment with new techniques and adapt them to their local contexts (Bakker et al., 2022; Friis-Hansen and Duveskog, 2012). By building on farmers' existing knowledge and experiences, participatory methods create a sense of ownership and empowerment, which are essential for sustaining innovation (Klerkx, 2020). The success of participatory approaches can be attributed to several factors. First, they recognize farmers as active agents in the innovation process rather than passive recipients of information (Hermans et al., 2021). This shift in perspective allows for the co-creation of knowledge and the development of context-specific solutions. Second, participatory approaches foster social learning and collaboration among farmers, enabling the rapid diffusion of innovative practices within farming communities (Knickel et al., 2009). Finally, these approaches are often more responsive to local needs and constraints, increasing the relevance and applicability of the innovations being promoted (Dolinska and d'Aquino, 2016). The findings of this review align with a growing body of literature that emphasizes the importance of participatory approaches in agricultural extension. Studies have shown that when farmers actively engage in the innovation process, they are more likely to adopt and adapt new practices to their specific circumstances (Tambo and Wünscher, 2018; Waters-Bayer et al., 2008). Moreover, innovations developed through participatory processes tend to be more sustainable and scalable, as they are tailored to the needs and capacities of local communities (Ensor and de Bruin, 2022).

5.2. Contextual factors influence effectiveness

The review also highlights the critical role of contextual factors in determining the effectiveness of different extension approaches

in enhancing farmers' innovativeness. Factors such as socio-economic conditions, environmental contexts, and institutional support can significantly influence the success of these interventions (Barzola Iza et al., 2020; Birner et al., 2009). For instance, studies have shown that farmers operating in supportive environments characterized by strong local networks and institutional backing are more likely to adopt innovative practices (Klerkx et al., 2012). In contrast, in less supportive contexts, even well-designed extension programs may struggle to achieve their intended outcomes. This suggests that a one-size-fits-all approach to extension is unlikely to be effective and that interventions must be tailored to the specific needs and capacities of local farming communities (Dolinska and d'Aquino, 2016).

Moreover, the review indicates that the effectiveness of extension approaches can vary significantly even within the same context. For example, Barzola Iza et al. (2020) found that the success of multi-stakeholder approaches, such as innovation platforms, depends heavily on local conditions. This variability highlights the need for ongoing monitoring and evaluation of extension programs to ensure that they remain relevant and effective in changing circumstances (Ensor and de Bruin, 2022). Understanding the role of contextual factors in influencing the effectiveness of extension approaches is essential for designing interventions that are both scalable and sustainable. By considering the specific socio-economic, environmental, and institutional contexts of farming communities, policy-makers and practitioners can develop more targeted and effective extension strategies that enhance farmers' innovativeness and promote sustainable agricultural practices (Klerkx, 2020).

5.3. Limitations and future research directions

While this systematic review provides valuable insights into the variability of innovation outcomes associated with different agricultural extension approaches, it also highlights several limitations and gaps in the existing literature. The review is based on a relatively small number of studies, many of which are context-specific and may not be generalizable to other settings. More research is needed to validate the findings and explore the effectiveness of extension approaches in a wider range of contexts. Additionally, longitudinal studies are essential for understanding the trajectories of innovation and the factors that contribute to the sustained adoption of new practices over time (Läpple et al., 2015). Another limitation of this study is its reliance primarily on a quantitative approach to compare the effectiveness of extension approaches, which may overlook the rich, context-specific insights that qualitative data can offer. More research is needed to unpack the complex relationships between extension methods, contextual factors, and innovation outcomes, and to identify the key drivers of successful innovation in agricultural systems (Salember et al., 2021). Despite these limitations, this systematic review provides a valuable starting point for understanding the variability of innovation outcomes associated with different agricultural extension approaches. By highlighting the importance of participatory, digital-based approaches and contextual factors in enhancing farmers' innovativeness, this review offers important insights for policymakers and practitioners seeking to promote sustainable agricultural practices and empower farmers to innovate in the face of complex challenges.

6. Conclusion

This systematic literature review has provided valuable insights into diverse agricultural extension approaches and their impact on farmers' innovativeness in Africa. The findings indicate that while various extension methods exist, their effectiveness in fostering innovation among farmers varies significantly. Participatory approaches, such as Farmer Field Schools and digital-based initiatives, have shown considerable promise in engaging farmers and enhancing their innovativeness. These approaches not only facilitate the transfer of knowledge but also empower farmers to adapt and develop solutions tailored to their specific contexts.

This review highlights the critical importance of contextual factors, including socio-economic conditions and institutional support, in determining the success of extension approaches. Interventions must be tailored to the unique circumstances of farming communities to maximize their impact. Furthermore, the findings underscore the need to shift from traditional top-down extension methods to more collaborative and farmer-centric approaches that recognize and build upon local knowledge and experiences.

In light of these insights, several recommendations can be made for policymakers and practitioners. First, enhancing digital extension services can significantly improve knowledge dissemination and decision-making among farmers, particularly in rural areas. Second, there is a need to prioritize participatory and integrated extension approaches that involve farmers in learning and innovation processes.

Despite the contributions of this review, gaps remain in the literature regarding the systematic evaluation of the impacts of different extension approaches on farmers' innovativeness. Future research should focus on longitudinal studies that assess the long-term effects of these methods and explore the interplay between contextual factors and innovation outcomes. By addressing these gaps, we can develop more effective agricultural policies and programs that support farmers in navigating the complex challenges of agricultural development.

Ethics statement

Not applicable because this work does not involve the use of animal or human subjects.

Declaration of competing interest

The authors hereby declare that there are no conflicts of interest related to this publication.

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References

- Afranaa Kwapong, N., Ankrah, D.A., 2023. Understanding innovation process within an interactive social network: empirical insights from maize innovations in southern Ghana. *Cogent Soc. Sci.* 9 (1). <https://doi.org/10.1080/23311886.2023.2167390>.
- Alaie, S.A., 2020. Knowledge and learning in the horticultural innovation system: a case of Kashmir valley of India. *Int. J. Innov. Stud.* 4 (4), 116–133. <https://doi.org/10.1016/j.ijis.2020.06.002>.
- Ariza, C., Rugeles, L., Saavedra, D., Guaitero, B., 2013. Measuring innovation in agricultural firms: a methodological approach. *Electron. J. Knowl. Manag.* 11, 185–198.
- Asare-Nuamah, P., Botchway, E., Onumah, J.A., 2019. Helping the helpless: contribution of rural extension services to smallholder farmers' climate change adaptive capacity and adaptation in rural Ghana. *Int. J. Rural Manag.* 15 (2), 244–268. <https://doi.org/10.1177/0973005219876211>.
- Bakker, T., Dugué, P., de Tourdonnet, S., 2021. Assessing the effects of Farmer Field Schools on farmers' trajectories of change in practices. *Agron. Sustain. Dev.* 41 (2), 18. <https://doi.org/10.1007/s13593-021-00667-2>.
- Bakker, T., Dugué, P., de Tourdonnet, S., 2022. How do farmers change their practices at the farm level after co-design processes in Farmer Field Schools? *Agric. Syst.* 201, 103457. <https://doi.org/10.1016/j.agry.2022.103457>.
- Barzola Iza, C.L., Dentoni, D., Omta, O.S.W.F., 2020. The influence of multi-stakeholder platforms on farmers' innovation and rural development in emerging economies: a systematic literature review. *J. Agribus. Dev. Emerg. Econ.* 10 (1), 13–39. <https://doi.org/10.1108/JADEE-12-2018-0182>.
- Bentley, J., Van Mele, P., Okry, F., Zossou, E., 2014. Videos that speak for themselves: when non-extensionists show agricultural videos to large audiences. *Dev. Pract.* 24 (7), 921–929. <https://doi.org/10.1080/09614524.2014.942216>.
- Berrueta, C., Giménez, G., Dogliotti, S., 2021. Scaling up from crop to farm level: Co-innovation framework to improve vegetable farm systems sustainability. *Agric. Syst.* 189, 103055. <https://doi.org/10.1016/j.agry.2021.103055>.
- Bhatta, G.D., Ojha, H.R., Aggarwal, P.K., Sulaiman, V.R., Sultana, P., Thapa, D., Mittal, N., Dahal, K., Thomson, P., Ghimire, L., 2017. Agricultural innovation and adaptation to climate change: empirical evidence from diverse agro-ecologies in South Asia. *Environ. Dev. Sustain.* 19 (2), 497–525. <https://doi.org/10.1007/s10668-015-9743-x>.
- Birner, R., Davis, K., Pender, J., Nkonya, E., Anandajayasekeram, P., Ekboir, J., Mbabu, A., Spielman, D.J., Horna, D., Benin, S., Cohen, M., 2009. From best practice to best fit: a framework for designing and analyzing pluralistic agricultural advisory services worldwide. *J. Agric. Educ. Ext.* 15 (4), 341–355. <https://doi.org/10.1080/13892240903309595>.
- Boulestreau, Y., Peyras, C.-L., Casagrande, M., Navarrete, M., 2022. Tracking down coupled innovations supporting agroecological vegetable crop protection to foster sustainability transition of agrifood systems. *Agric. Syst.* 196, 103354. <https://doi.org/10.1016/j.agry.2021.103354>.
- Brown, P., Roper, S., 2017. Innovation and networks in New Zealand farming. *Aust. J. Agric. Resour. Econ.* 61 (3), 422–442. <https://doi.org/10.1111/1467-8489.12211>.
- Brun, J., Jeuffroy, M.-H., Pénicaud, C., Cerf, M., Meynard, J.-M., 2021. Designing a research agenda for coupled innovation towards sustainable agrifood systems. *Agric. Syst.* 191, 103143. <https://doi.org/10.1016/j.agry.2021.103143>.
- Colnago, P., Rossing, W.A.H., Dogliotti, S., 2021. Closing sustainability gaps on family farms: combining on-farm co-innovation and model-based explorations. *Agric. Syst.* 188, 103017. <https://doi.org/10.1016/j.agry.2020.103017>.
- Daniso, B., Muche, M., Fikadu, B., Melaku, E., Lemma, T., 2020. Assessment of rural households' mobile phone usage status for rural innovation services in Gomma Woreda, Southwest Ethiopia. *Cogent Food Agric.* 6 (1), 1728083. <https://doi.org/10.1080/23311932.2020.1728083>.
- Davis, K., Rosenbach, G., Spielman, D.J., Makhija, S., Mwangi, L., 2024. Assessing Agricultural Extension Agent Digital Readiness in Rwanda. In: *Intl Food Policy Res Inst.*
- Dayamba, D.S., Ky-Dembele, C., Bayala, J., Dorward, P., Clarkson, G., Sanogo, D., Diop Mamadou, L., Traoré, I., Diakité, A., Nenkam, A., Binam, J.N., Ouedraogo, M., Zougmore, R., 2018. Assessment of the use of Participatory Integrated Climate Services for Agriculture (PICSA) approach by farmers to manage climate risk in Mali and Senegal. *Climate Services* 12, 27–35. <https://doi.org/10.1016/j.cliser.2018.07.003>.
- Diederer, P., Meijl, H., Van, Wolters, A., Bijak, K., Van Meijl, H., 2003. Innovation adoption in agriculture : innovators, early adopters and laggards innovation adoption in agriculture: innovators, early adopters and laggards. *Cah. Econ. Sociol. Rurales* 29–50, 2003. <https://hal.science/hal-01201041>.
- Dolinska, A., d'Aquino, P., 2016. Farmers as agents in innovation systems. Empowering farmers for innovation through communities of practice. *Agric. Syst.* 142, 122–130. <https://doi.org/10.1016/j.agry.2015.11.009>.
- Dolinska, A., Oates, N., Ludi, E., Habtu, S., Rougier, J., Sanchez-Reparaz, M., Mosello, B., Yazew, E., Kifle, M., Fissehay, D., Aregay, G., Tamele, H.F., Barberá, G.G., d'Aquino, P., 2020. Engaging farmers in a research project. Lessons learned from implementing the community of practice concept in innovation platforms in irrigated schemes in Tunisia, Mozambique and Ethiopia. *Irrig. Drain.* 69 (S1), 38–48. <https://doi.org/10.1002/ird.2222>.
- Dosso, F., Gouroubera, M.W., Idrissou, L., Moumouni-Moussa, I., 2023. The combination of extension approaches strengthens farmers' innovativeness in sustainable land management. *Environ. Dev. Sustain.* 26 (4), 10043–10062. <https://doi.org/10.1007/s10668-023-03134-y>.
- Ensor, J., de Bruin, A., 2022. The role of learning in farmer-led innovation. *Agric. Syst.* 197, 103356. <https://doi.org/10.1016/j.agry.2021.103356>.
- Fa, A., 2018. Appraisal of innovations in agricultural extension and advisory services in Cameroon. In: Citation: Amungwa FA, 1. www.scholarena.com.
- Fagerberg, J., Mowery, C.D., Nelson, R.R., 2005. *The Oxford Handbook of Innovation*. Oxford University Press.
- Faturoti, B.O., Agwu, A.E., Igboke, E.M., Tenkouano, A., 2008. International institute of tropical agriculture plantain and banana programme: an insight into the contributions of farmer-to-farmer extension paradigm. *Afr. J. Biotechnol.* 7 (13), 2137–2146. <http://www.academicjournals.org/AJB>.
- Faure, G., Penot, E., Rakotondravelo, J.C., Ramahatoraka, H.A., Dugué, P., Toillier, A., 2013. Which advisory system to support innovation in conservation agriculture? The case of Madagascar's lake Alaotra. *J. Agric. Educ. Ext.* 19 (3), 257–270. <https://doi.org/10.1080/1389224X.2013.782169>.
- Fayse, N., Sraïri, M.T., Errahj, M., 2012. Local farmers' organisations: a space for peer-to-peer learning? The case of milk collection cooperatives in Morocco. *J. Agric. Educ. Ext.* 18 (3), 285–299. <https://doi.org/10.1080/1389224X.2012.670053>.
- Feng, H., 2024. The evolution and potential of precision agriculture in China anchored in “3S” technology. *Highlights Sci. Eng. Technol.* 81, 592–597. <https://doi.org/10.54097/43n29y21>.
- Friis-Hansen, E., Duveskog, D., 2012. The empowerment route to well-being: an analysis of farmer field schools in East Africa. *World Dev.* 40 (2), 414–427. <https://doi.org/10.1016/j.worlddev.2011.05.005>.
- Fuentes, R.A., Soto, A.R., 2015. Non-technological innovations in Chilean agricultural firms: what motivates the decision to innovate and the propensity of innovation? *Cienc. Investig. Agrar.* 42 (2). <https://doi.org/10.4067/S0718-16202015000200004>, 4–4.
- Gachanja, I.M., 2023. Influence of technology transfer from universities on manufacturing firms' innovative performance. *Int. J. Innov. Stud.* 7 (2), 115–126. <https://doi.org/10.1016/j.ijis.2022.12.003>.
- Gault, F., 2013. *The Oslo manual*. In: *Handbook of Innovation Indicators and Measurement*. Edward Elgar.
- Gouroubera, M.W., Adechian, S.A., Segnon, A.C., Moumouni-Moussa, I., Zougmore, R.B., 2025. Drivers and impacts of mobile phone-mediated scaling of agricultural technologies: a meta-analysis. *Front. Sustain. Food Syst.* 8. <https://doi.org/10.3389/fsufs.2024.1514546>.
- Gouroubera, M.W., Idrissou, L., Moumouni, I.M., Okry, F., Baco, M.N., 2023a. Institutionalization as an innovation process: insight into ICT use in agricultural advisory system. *Int. J. Innovat. Technol. Manag.* 20 (2). <https://doi.org/10.1142/S021987702330001X>.

- Gouroubra, M.W., Kora Sabi, A., Bio Comada, T.K., Dosso, F., Fatondji, S.A., Gouthon, M.B., Houaga, R.P., 2023b. Designing effective digital-based delivery of climate information for smallholder farmers: a mini meta-analysis on drivers and barriers. *Clim. Policy* 1–14. <https://doi.org/10.1080/14693062.2023.2266475>.
- Gouroubra, M.W., Moumouni, I.M., Okry, F., Idrissou, L., 2023c. A holistic approach to understanding ICT implementation challenges in rural advisory services: lessons from using farmer learning videos. *J. Agric. Educ. Ext.* 1–20. <https://doi.org/10.1080/1389224X.2023.2171077>.
- Gunnarsson, J., 2021. Pathways to innovation for Gambian farmers through supplier contracts. *Sci. Afr.* 14, e00990. <https://doi.org/10.1016/j.sciaf.2021.e00990>.
- Hang, C.C., Chen, J., 2021. Innovation management research in the context of developing countries: analyzing the disruptive innovation framework. *Int. J. Innov. Stud.* 5 (4), 145–147. <https://doi.org/10.1016/j.ijis.2021.09.001>.
- Hermans, T.D.G., Smith, H.E., Whitfield, S., Sallu, S.M., Recha, J., Dougill, A.J., Thierfelder, C., Gama, M., Bunderson, W.T., Museka, R., Daggart, N., Meshack, C., 2023. Role of the interaction space in shaping innovation for sustainable agriculture: empirical insights from African case studies. *J. Rural Stud.* 100, 103012. <https://doi.org/10.1016/j.jrurstud.2023.103012>.
- Hermans, T.D.G., Whitfield, S., Dougill, A.J., Thierfelder, C., 2021. Why we should rethink ‘adoption’ in agricultural innovation: empirical insights from Malawi. *Land Degrad. Dev.* 32 (4), 1809–1820. <https://doi.org/10.1002/ldr.3833>.
- Karubanga, G., Kibwika, P., Okry, F., Sseguya, H., 2016. Empowering farmers to learn and innovate through integration of video-mediated and face-to-face extension approaches: the case of rice farmers in Uganda. *Cogent Food Agric.* 2 (1). <https://doi.org/10.1080/23311932.2016.1274944>.
- Karubanga, G., Kibwika, P., Okry, F., Sseguya, H., 2017. How farmer videos trigger social learning to enhance innovation among smallholder rice farmers in Uganda. *Cogent Food Agric.* 3 (1), 1368105. <https://doi.org/10.1080/23311932.2017.1368105>.
- Kiptot, E., Franzel, S., 2015. Farmer-to-farmer extension: opportunities for enhancing performance of volunteer farmer trainers in Kenya. *Dev. Pract.* 25 (4), 503–517. <https://doi.org/10.1080/09614524.2015.1029438>.
- Klerkx, L., 2020. Advisory services and transformation, plurality and disruption of agriculture and food systems: towards a new research agenda for agricultural education and extension studies. *J. Agric. Educ. Ext.* 26 (2), 131–140. <https://doi.org/10.1080/1389224X.2020.1738046>.
- Klerkx, L., 2021. Digital and virtual spaces as sites of extension and advisory services research: social media, gaming, and digitally integrated and augmented advice. *J. Agric. Educ. Ext.* 27 (3), 277–286. <https://doi.org/10.1080/1389224X.2021.1934998>.
- Klerkx, L., van Mierlo, B., Leeuwis, C., 2012. Evolution of systems approaches to agricultural innovation: concepts, analysis and interventions. In: *Farming Systems Research into the 21st Century: the New Dynamic*. Springer, Netherlands, pp. 457–483. https://doi.org/10.1007/978-94-007-4503-2_20.
- Knickel, K., Brunori, G., Rand, S., Proost, J., 2009. Towards a better conceptual framework for innovation processes in agriculture and rural development: from linear models to systemic approaches. *J. Agric. Educ. Ext.* 15 (2), 131–146. <https://doi.org/10.1080/13892240902909064>.
- Läpple, D., Renwick, A., Thorne, F., 2015. Measuring and understanding the drivers of agricultural innovation: evidence from Ireland. *Food Policy* 51, 1–8. <https://doi.org/10.1016/j.foodpol.2014.11.003>.
- Lawal, B.O., Akintayo, O.I., Ayoola, O.T., Oyedokun, M.O., Taiwo, L.B., Oyedele, D.J., Adebooye, O.C., 2019. scp>F</scp>armers’ participation in vegetable innovation platforms in Southwest Nigeria. *Acta Hort.* 1238, 157–164. <https://doi.org/10.17660/ActaHortic.2019.1238.16>.
- Lema, Z., Lobry de Bruyn, L.A., Marshall, G.R., Roschinsky, R., Duncan, A.J., 2021. Multilevel innovation platforms for development of smallholder livestock systems: how effective are they? *Agric. Syst.* 189, 103047. <https://doi.org/10.1016/j.agsy.2020.103047>.
- Maryono, M., Kiloes, A.M., Adhikari, R., Abdul Aziz, A., 2024. Agriculture development through multi-stakeholder partnerships in developing countries: a systematic literature review. *Agric. Syst.* 213, 103792. <https://doi.org/10.1016/j.agsy.2023.103792>.
- Meynard, J.-M., Jeuffroy, M.-H., Le Bail, M., Lefevre, A., Magrini, M.-B., Michon, C., 2017. Designing coupled innovations for the sustainability transition of agrifood systems. *Agric. Syst.* 157, 330–339. <https://doi.org/10.1016/j.agsy.2016.08.002>.
- Moher, D., 2009. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *Ann. Intern. Med.* 151 (4), 264. <https://doi.org/10.7326/0003-4819-151-4-200908180-00135>.
- Munthali, N., Leeuwis, C., van Paassen, A., Lie, R., Asare, R., van Lammeren, R., Schut, M., 2018. Innovation intermediation in a digital age: comparing public and private new-ICT platforms for agricultural extension in Ghana. *NJAS - Wageningen J. Life Sci.* 86–87 (1), 64–76. <https://doi.org/10.1016/j.njas.2018.05.001>.
- Naika, M.B.N., Kudari, M., Devi, M.S., Sadhu, D.S., Sunagar, S., 2021. Digital extension service. In: *Food Technology Disruptions*. Elsevier, pp. 285–323. <https://doi.org/10.1016/B978-0-12-821470-1.00006-9>.
- Nakano, Y., Tsusaka, T.W., Aida, T., Pede, V.O., 2018. Is farmer-to-farmer extension effective? The impact of training on technology adoption and rice farming productivity in Tanzania. *World Dev.* 105, 336–351. <https://doi.org/10.1016/j.worlddev.2017.12.013>.
- OECD, 2013. *Agricultural Innovation Systems*. OECD. <https://doi.org/10.1787/9789264200593-en>.
- Ogisi, O.D., Begho, T., 2023. Adoption of climate-smart agricultural practices in sub-Saharan Africa: a review of the progress, barriers, gender differences and recommendations. *Farming Syst.* 1 (2), 100019. <https://doi.org/10.1016/j.farsys.2023.100019>.
- Okori, P., Munthali, W., Msere, H., Charlie, H., Chitaya, S., Sichali, F., Chilumpha, E., Chirwa, R., Seetha, A., Chinyamuyamu, B., Monyo, E., Siambi, M., Chirwa, R., 2022. Improving efficiency of knowledge and technology diffusion using community seed banks and farmer-to-farmer extension: experiences from Malawi. *Agric. Food Secur.* 11 (1), 38. <https://doi.org/10.1186/s40066-022-00375-4>.
- Osumba, J.J.L., Recha, J.W., Oroma, G.W., 2021. Transforming agricultural extension service delivery through innovative bottom-up climate-resilient agribusiness farmer field schools. *Sustainability* 13 (7), 3938. <https://doi.org/10.3390/su13073938>.
- Parry, K., van Rooyen, A.F., Bjornlund, H., Kissoly, L., Moyo, M., de Sousa, W., 2020. The importance of learning processes in transitioning small-scale irrigation schemes. *Int. J. Water Resour. Dev.* 36 (Suppl. 1), S199–S223. <https://doi.org/10.1080/07900627.2020.1767542>.
- Pawera, L., Manickam, R., Wangungu, C., Bonnath, U., Schreinemachers, P., Ramasamy, S., 2024. Guidance on farmer participation in the design, testing and scaling of agricultural innovations. *Agric. Syst.* 218, 104006. <https://doi.org/10.1016/j.agsy.2024.104006>.
- Razafinimpiasa, L., Rabeharisoa, L., Temple, L., 2022. Chemins d’impacts de la recherche agronomique appliquée à un projet de fertilisation phosphatée de la riziculture pluviale à Madagascar. *Tropicicultura*. <https://doi.org/10.25518/2295-8010.2046>.
- Rogers, E.M., Singhal, A., Quinlan, M.M., 2014. Diffusion of innovations. In: *An Integrated Approach to Communication Theory and Research*. Routledge.
- Salembier, C., Segrestin, B., Weil, B., Jeuffroy, M.-H., Cadoux, S., Cros, C., Favrelière, E., Fontaine, L., Gimaret, M., Noilhan, C., Petit, A., Petit, M.-S., Porhiet, J.-Y., Sicard, H., Reau, R., Ronceux, A., Meynard, J.-M., 2021. A theoretical framework for tracking farmers’ innovations to support farming system design. *Agron. Sustain. Dev.* 41 (5), 61. <https://doi.org/10.1007/s13593-021-00713-z>.
- Schut, M., Rodenburg, J., Klerkx, L., van Ast, A., Bastiaans, L., 2014. Systems approaches to innovation in crop protection. A systematic literature review. *Crop Prot.* 56, 98–108. <https://doi.org/10.1016/j.cropro.2013.11.017>.
- Spurk, C., Koch, C., Bürgin, R., Chikopela, L., Konaté, F., Nyabuga, G., Sarpong, D.B., Sousa, F., Fliessbach, A., 2023. Farmers’ innovativeness and positive affirmation as main drivers of adoption of soil fertility management practices – evidence across sites in Africa. *J. Agric. Educ. Ext.* 1–25. <https://doi.org/10.1080/1389224X.2023.2281909>.
- Tambo, J.A., Aliamo, C., Davis, T., Mugambi, I., Romney, D., Onyango, D.O., Kansime, M., Alokot, C., Byantwale, S.T., 2019. The impact of ICT-enabled extension campaign on farmers’ knowledge and management of fall armyworm in Uganda. *PLoS One* 14 (8), e0220844. <https://doi.org/10.1371/journal.pone.0220844>.
- Tambo, J.A., Wünscher, T., 2018. Building farmers’ capacity for innovation generation: insights from rural Ghana. *Renew. Agric. Food Syst.* 33 (2), 116–130. <https://doi.org/10.1017/S1742170516000521>.
- Turyahikayo, W., Matsiko, B.F., Okiror, J.J., Obaa, B.B., Hanf, J.H., 2019. Influence of social mechanisms on innovation behaviour of actors in agricultural innovation platforms in Uganda. *J. Agric. Ext.* 23 (3), 23. <https://doi.org/10.4314/jae.v23i3.2>.
- Walder, P., Sinabell, F., Unterlass, F., Niedermayr, A., Fulgeanu, D., Kapfer, M., Melcher, M., Kantelhardt, J., 2019. Exploring the relationship between farmers’ innovativeness and their values and aims. *Sustainability* 11 (20), 5571. <https://doi.org/10.3390/su11205571>.
- Waters-Bayer, A., Laurens, van V., Mariana, W., Chesha, W., 2008. *Recognizing and Enhancing Processes of Local Innovation*, first ed. Earthscan, London.
- Wellard, K., Rafanomezana, J., Nyirenda, M., Okoteli, M., Subbey, V., 2013. A review of community extension approaches to innovation for improved livelihoods in Ghana, Uganda and Malawi. *J. Agric. Educ. Ext.* 19 (1), 21–35. <https://doi.org/10.1080/1389224X.2012.714712>.

- Yao, S., Wang, Y., Wu, G., 2024. Measurement and influencing factor analysis of the matching degree between agricultural scientific and technology innovation and rural industry integration under the sustainable background. *Front. Sustain. Food Syst.* 8. <https://doi.org/10.3389/fsufs.2024.1362877>.
- Zossou, E., Van Mele, P., Vodouhe, S.D., Wanvoeke, J., 2009. The power of video to trigger innovation: rice processing in central Benin. *Int. J. Agric. Sustain.* 7 (2), 119–129. <https://doi.org/10.3763/ijas.2009.0438>.
- Zossou, E., Van Mele, P., Vodouhe, S.D., Wanvoeke, J., 2010. Women groups formed in response to public video screenings on rice processing in Benin. *Int. J. Agric. Sustain.* 8 (4), 270–277. <https://doi.org/10.3763/ijas.2010.0499>.
- Zoundji, G.C., Okry, F., Van Mele, P., Bentley, J.W., Kwame Sackey, C., 2024. The potential of farmer training video for supporting agroecological vegetable production in Benin. *Cogent Food Agric.* 10 (1). <https://doi.org/10.1080/23311932.2024.2358607>.